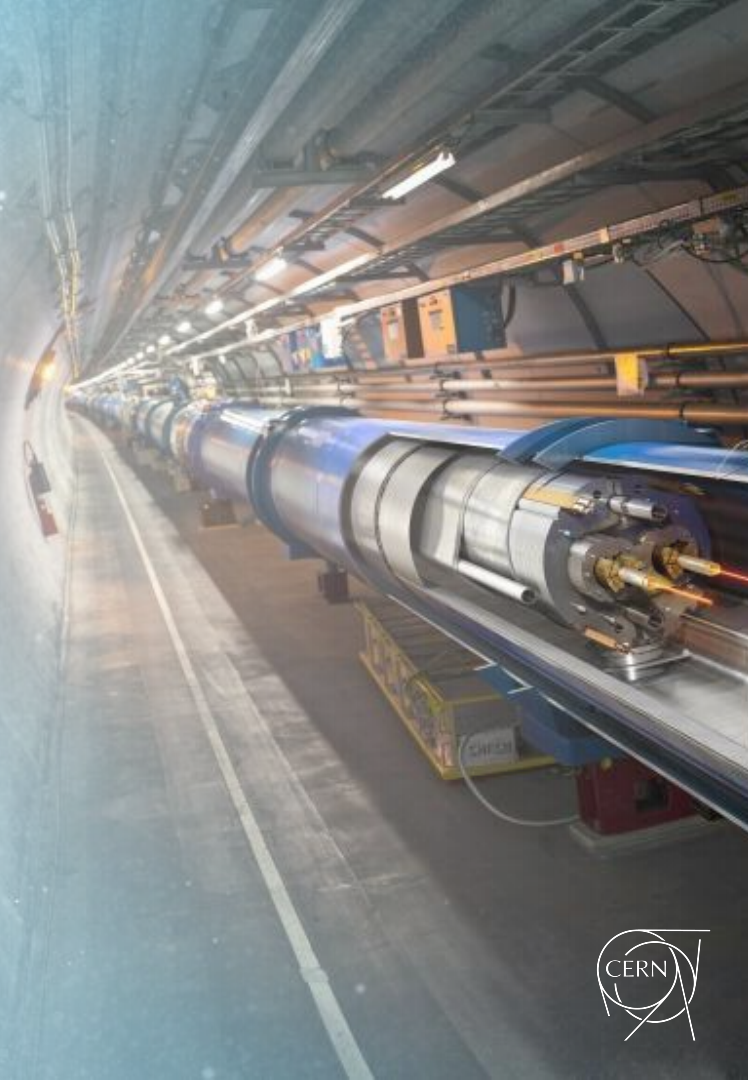
Kernel Recipes 2024

How CERN serves 1 EB of data via FUSE



G. Amadio
CERN IT Storage and Data Management

CERN



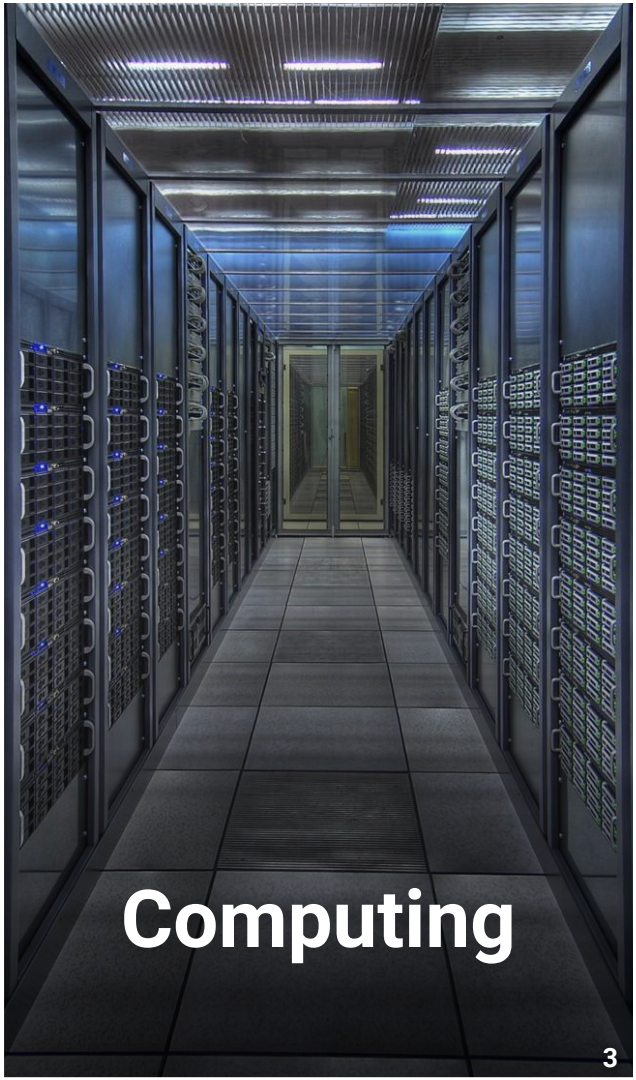
CERN,
the European Organisation
for Nuclear Research, is widely
recognised as one of the world's
leading laboratories for
fundamental research.

CERN is an
intergovernmental organisation
with 24 Member States and 10
Associate Member States.
It is situated on the French-Swiss
border, near Geneva.

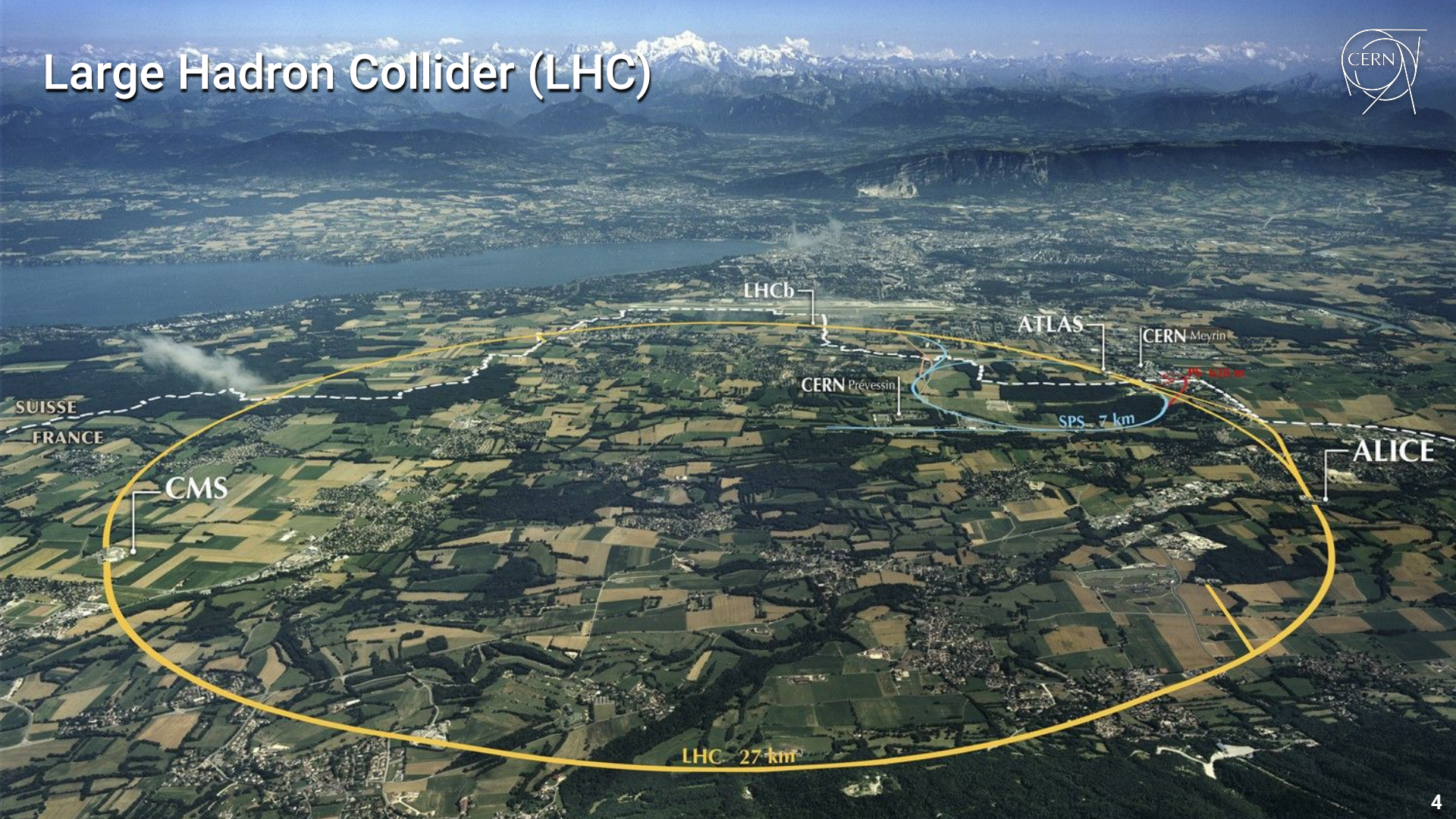
CERN's mission is
to gain understanding of
the most fundamental particles
and laws of the universe.


Accelerators

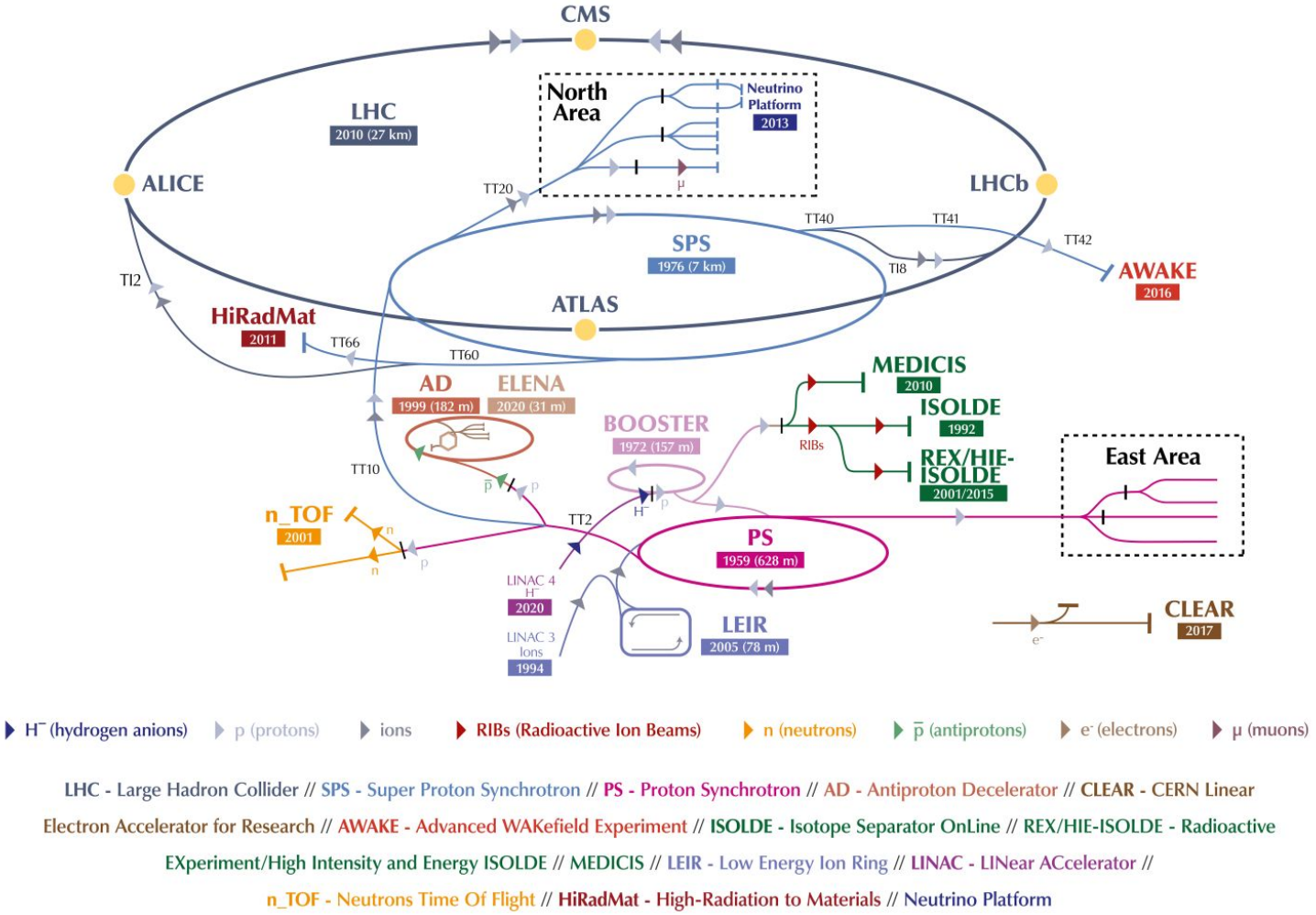

Detectors


Computing

Large Hadron Collider (LHC)

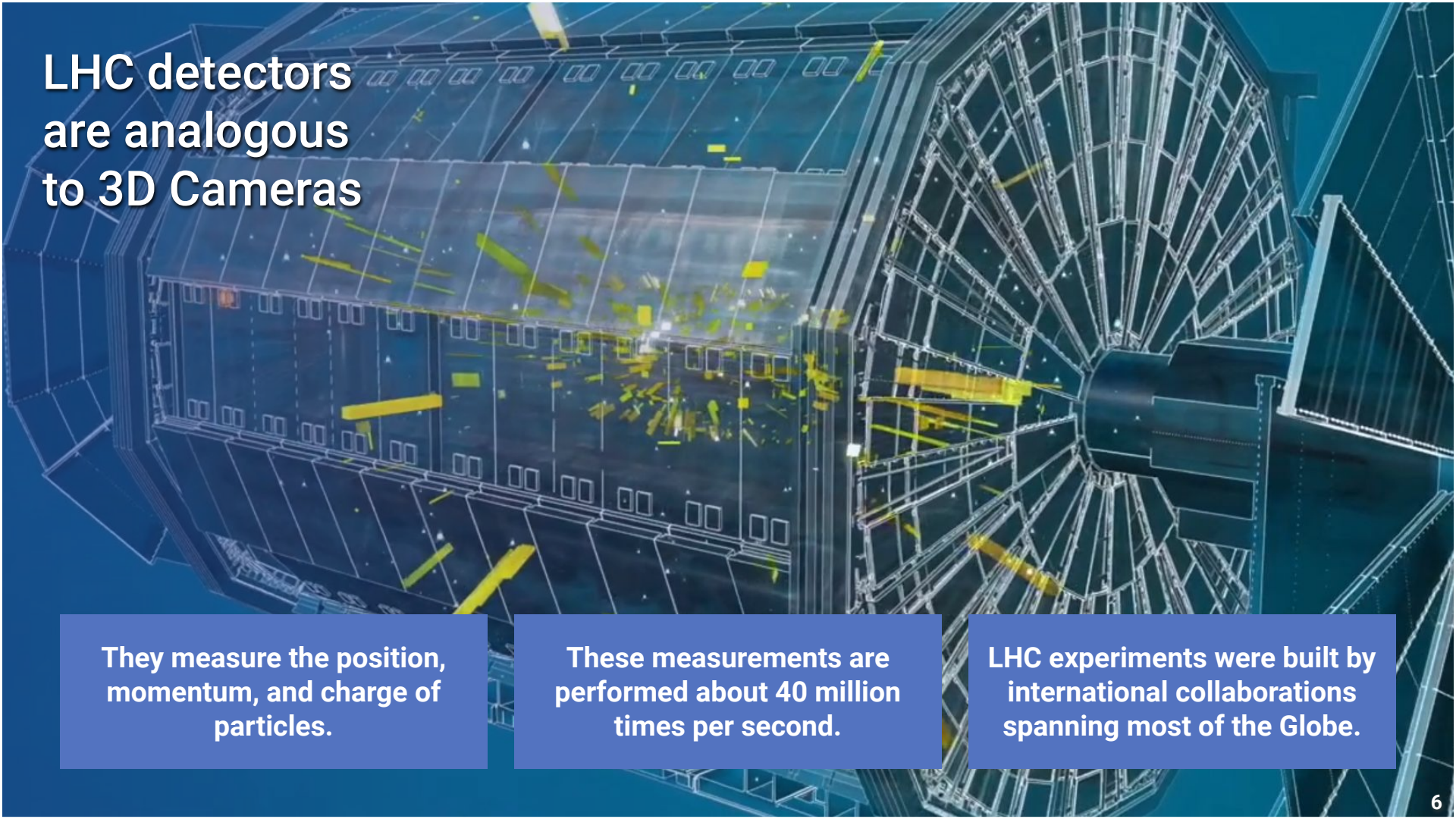


CERN Accelerator Complex



LHC - Large Hadron Collider // SPS - Super Proton Synchrotron // PS - Proton Synchrotron // AD - Antiproton Decelerator // CLEAR - CERN Linear Electron Accelerator for Research // AWAKE - Advanced WAKefield Experiment // ISOLDE - Isotope Separator OnLine // REX/HIE-ISOLDE - Radioactive EXperiment/High Intensity and Energy ISOLDE // MEDICIS // LEIR - Low Energy Ion Ring // LINAC - LINear ACcelerator // n_TOF - Neutrons Time Of Flight // HiRadMat - High-Radiation to Materials // Neutrino Platform

LHC detectors are analogous to 3D Cameras

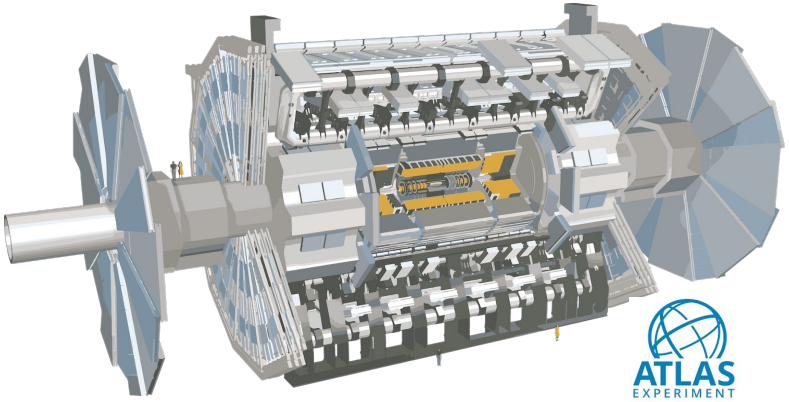


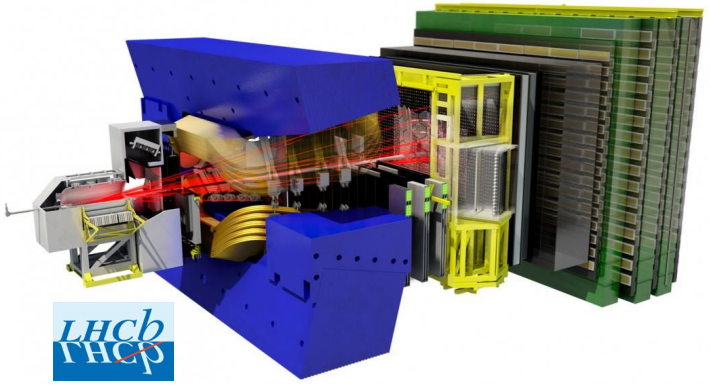
They measure the position, momentum, and charge of particles.

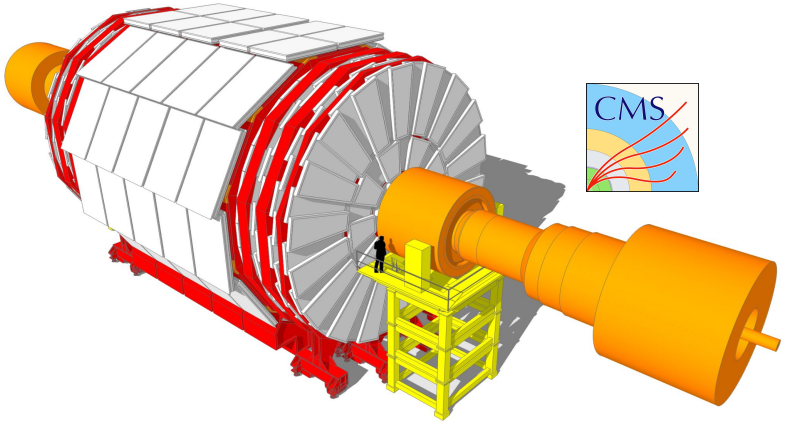
These measurements are performed about 40 million times per second.

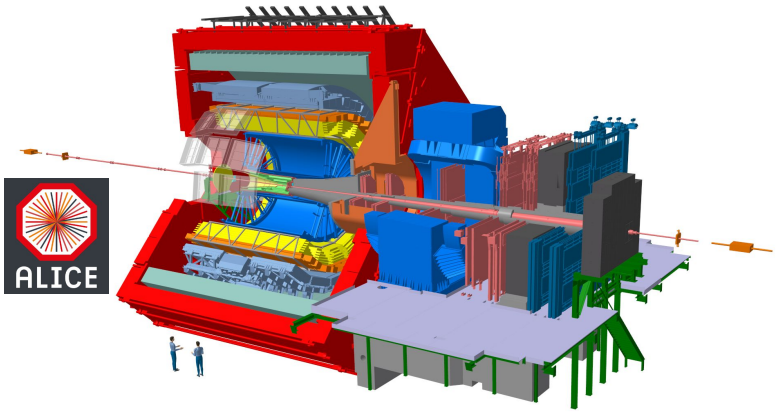
LHC experiments were built by international collaborations spanning most of the Globe.

Main Experiments at the LHC

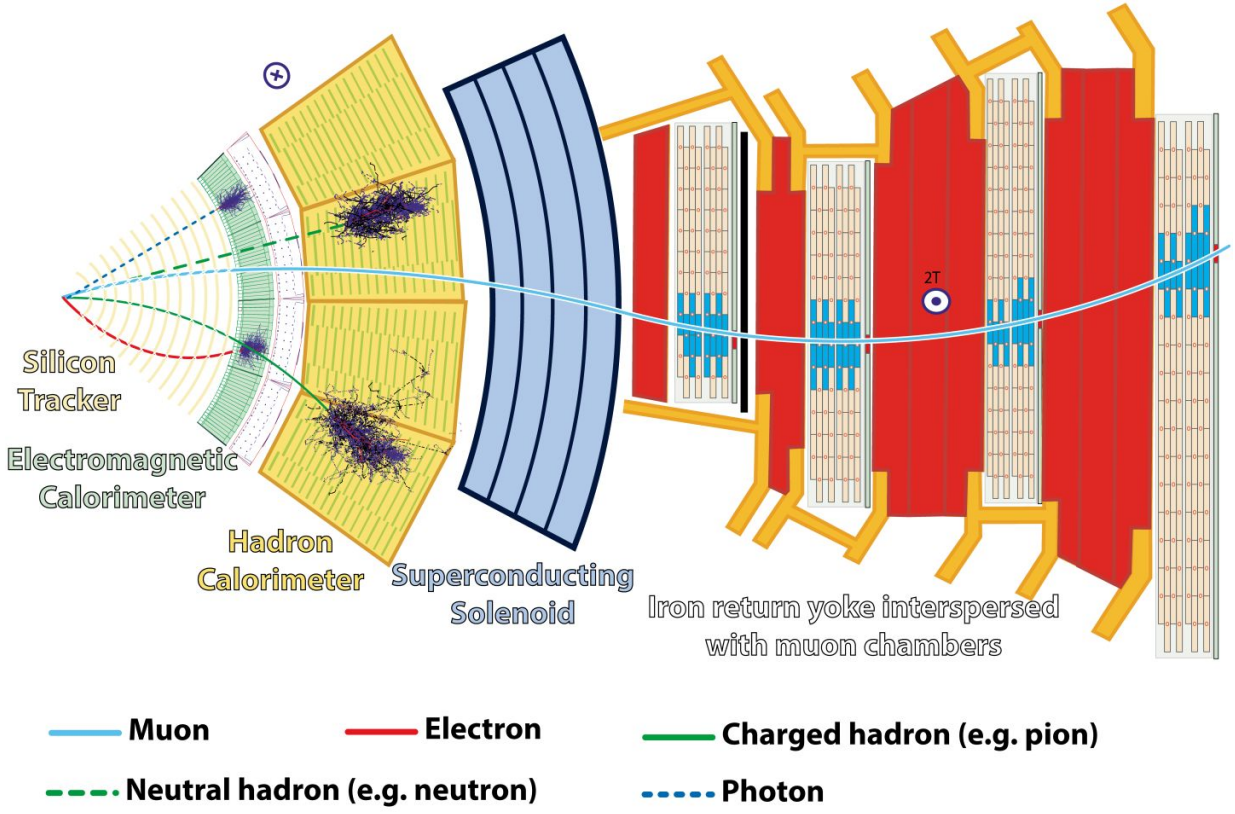








How Particles Interact



Deluge of Data

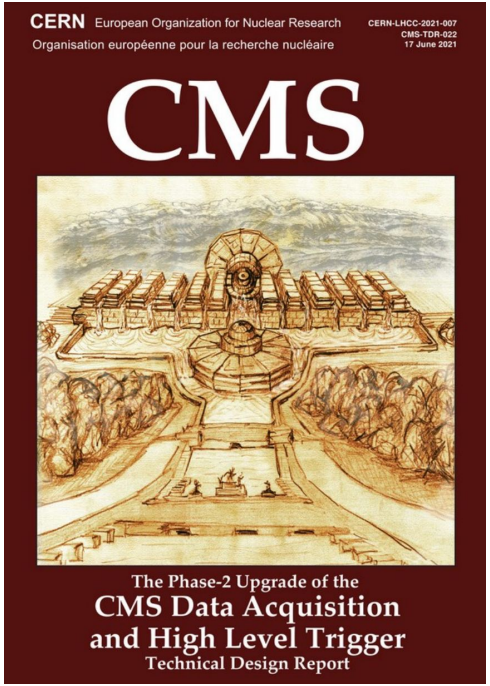
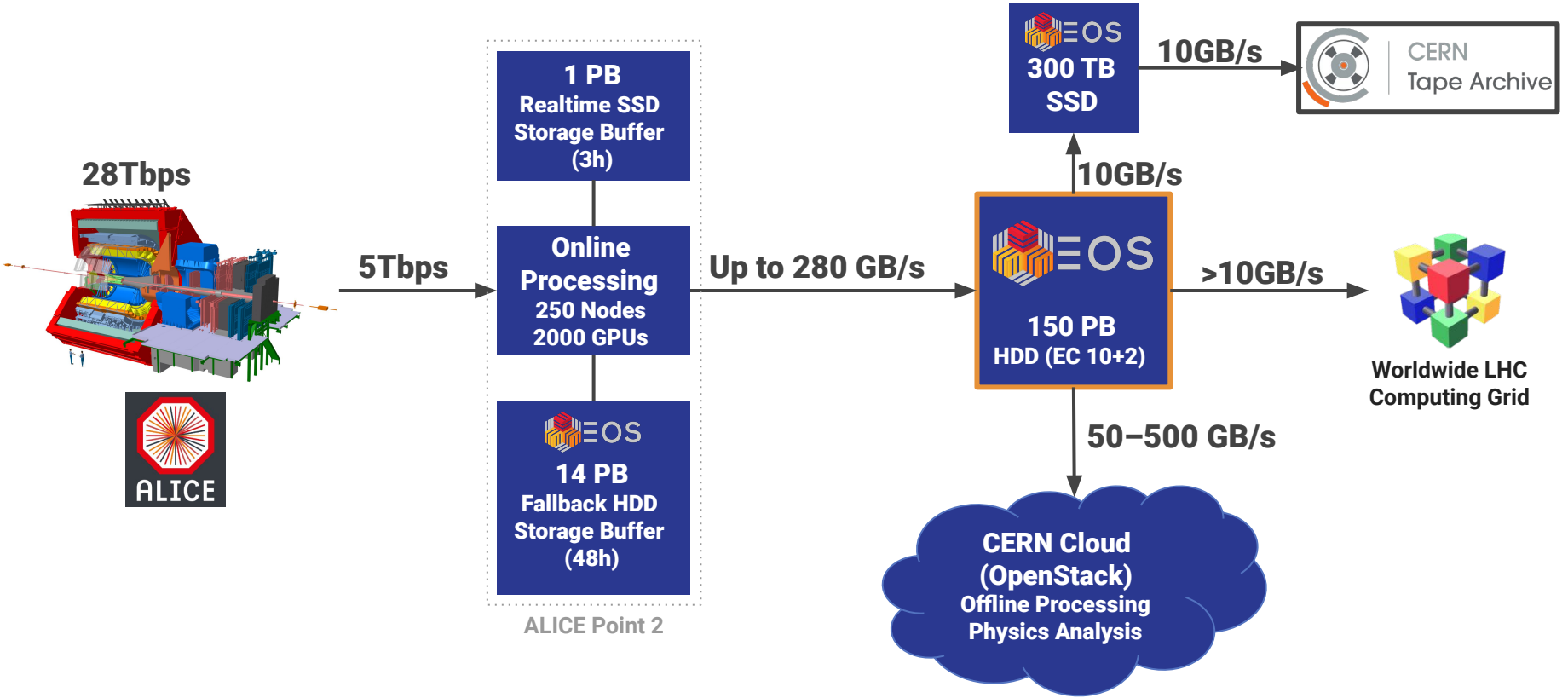


Table 1.2: CMS Phase-2 trigger and DAQ projected running parameters, compared to the design values of the current (Phase-1) system.

	LHC	HL-LHC	
CMS detector	Phase-1	Phase-2	
Peak ⟨PU⟩	60	140	200
L1 accept rate (maximum)	100 kHz	500 kHz	750 kHz
Event Size at HLT input	2.0 MB [a]	6.1 MB	8.4 MB
Event Network throughput	1.6 Tb/s	24 Tb/s	51 Tb/s
Event Network buffer (60 s)	12 TB	182 TB	379 TB
HLT accept rate	1 kHz	5 kHz	7.5 kHz
HLT computing power [b]	0.7 MHS06	17 MHS06	37 MHS06
Event Size at HLT output [c]	1.4 MB	4.3 MB	5.9 MB
Storage throughput [d]	2 GB/s	24 GB/s	51 GB/s
Storage throughput (Heavy-Ion)	12 GB/s	51 GB/s	51 GB/s
Storage capacity needed (1 day [e])	0.2 PB	1.6 PB	3.3 PB

[a] Design value.
[b] Does not include Data Quality Monitoring.
[c] Actual compression factor for Phase-1. For Phase-2 same factor is assumed, see Section 6.2.11.
[d] The storage throughput is defined as the effective throughput with concurrent recording and transfer. The throughput required is determined by the HLT output event size and the additional output streams, see Section 6.2.11.
[e] Assuming an LHC duty cycle, i.e. the fraction of time spent in stable colliding beams, of 75 %.

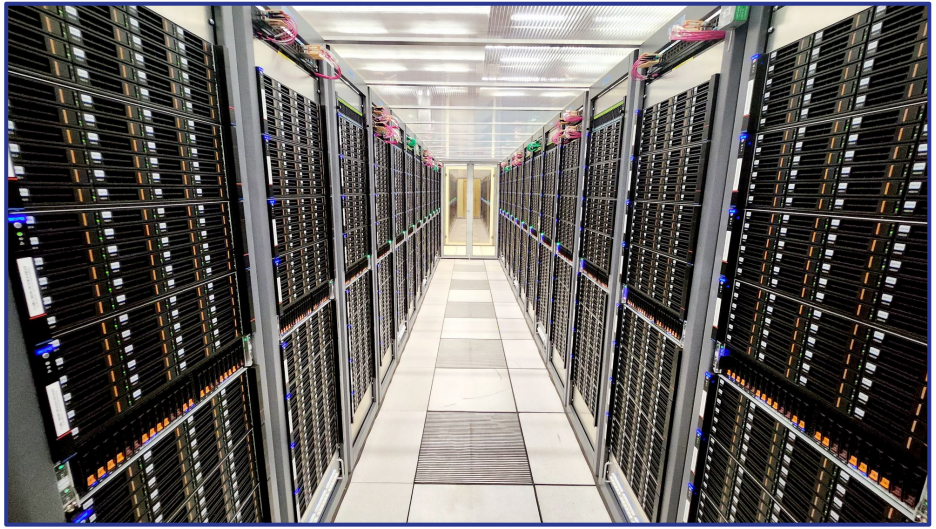
ALICE Experiment Data Flow (Heavy Ion, Run 3)



EOS ALICE O² Installation at CERN

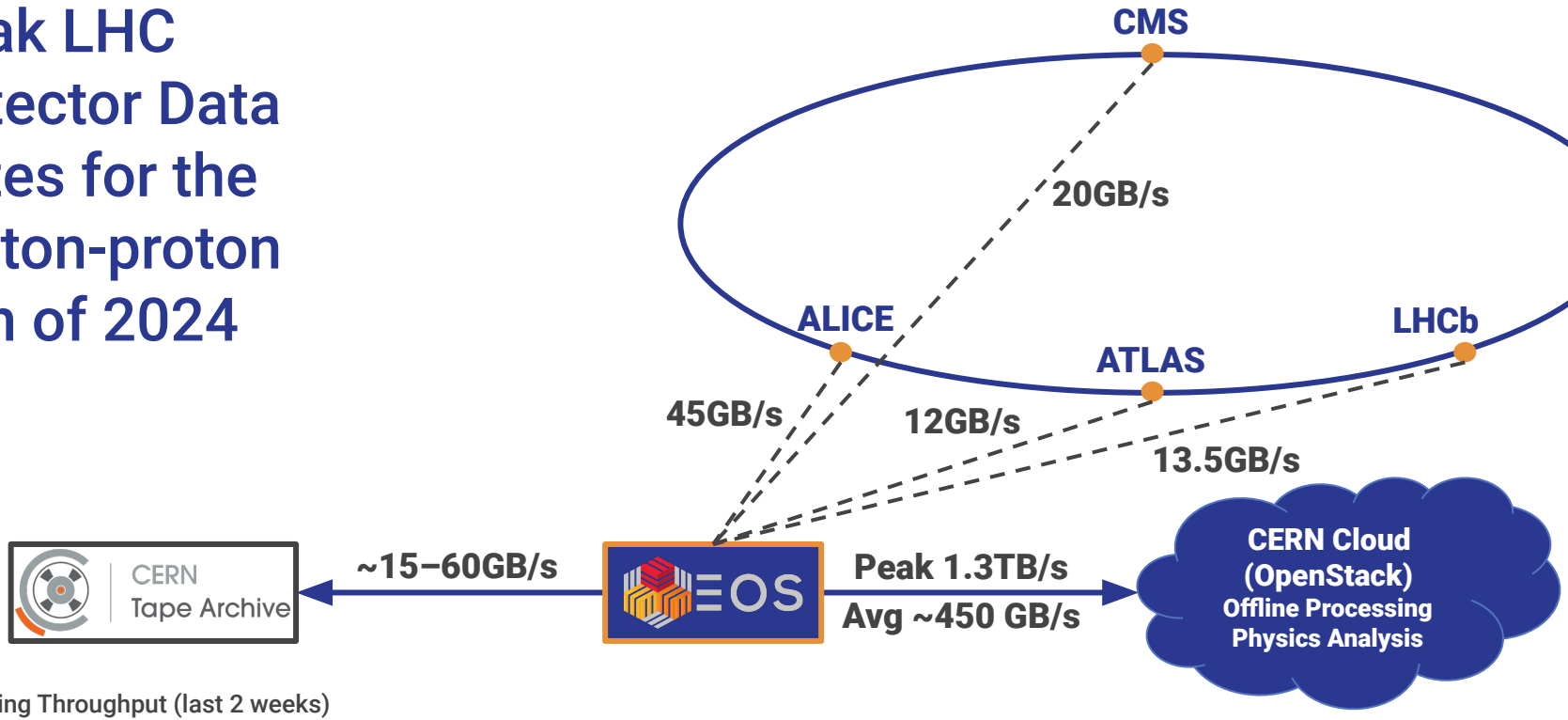
- 180 PB raw, 150 PB usable space
- Layout: 10 + 2 erasure coding
- 12000 HDDs in total
- About ~1/3 shown below (3840 HDDs)

- 1.3 TB/s peak read throughput
- 380 GB/s peak write throughput
- 96 x 18TB per node
- 100Gbps networking

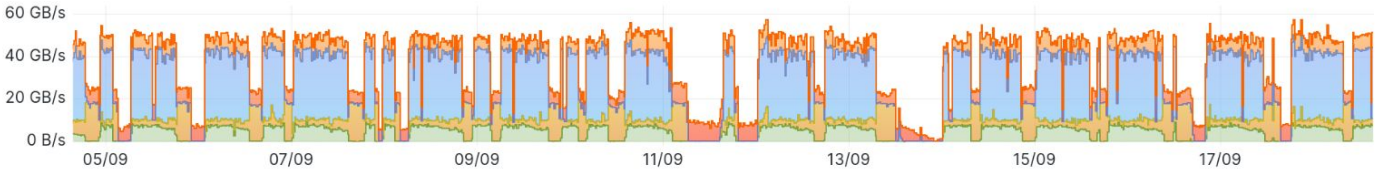




Peak LHC Detector Data Rates for the proton-proton Run of 2024

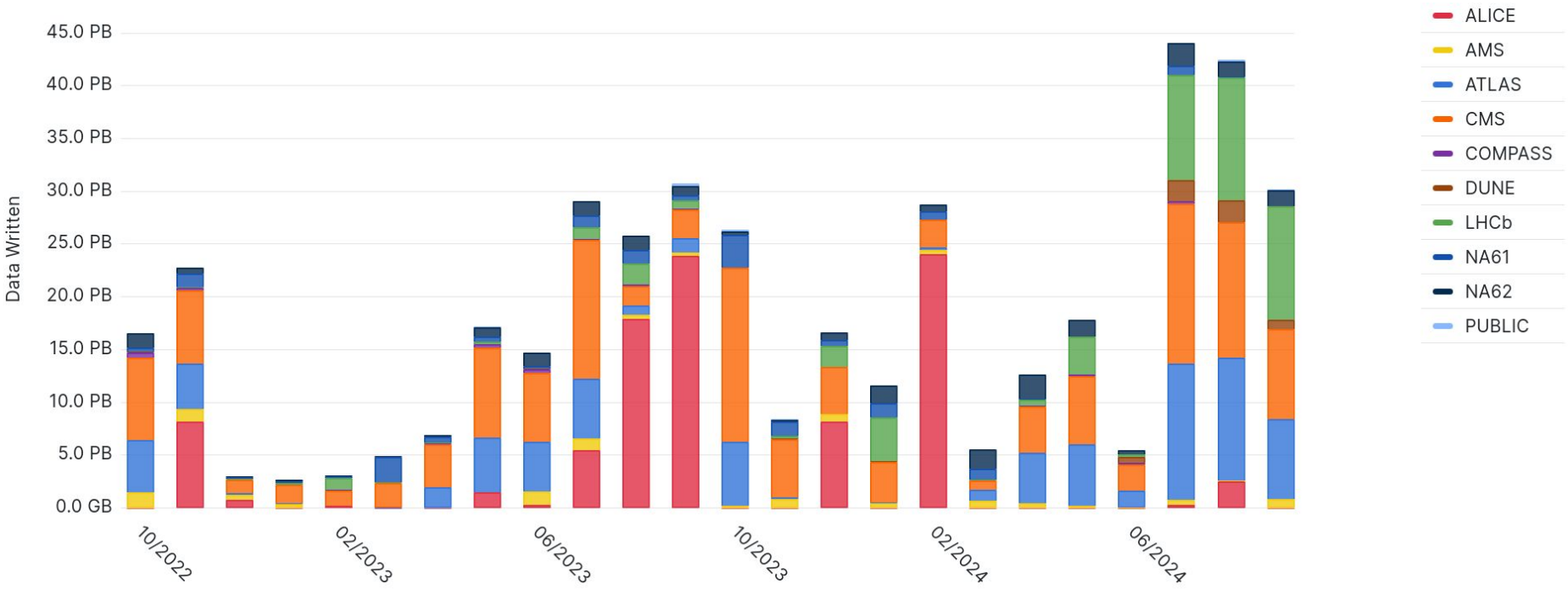


Data Taking Throughput (last 2 weeks)



Name	Max	Mean	Last *
ATLAS Point1	9.70 GB/s	4.43 GB/s	7.46 GB/s
CMS Point5	19.7 GB/s	5.92 GB/s	3.36 GB/s
ALICE O2 Point2	39.7 GB/s	20.6 GB/s	33.8 GB/s
LHCb Point8	12.1 GB/s	6.47 GB/s	6.92 GB/s

Data Volume to Tape (Last 2 Years)



An exabyte of disk storage at CERN



ABOUT NEWS SCI

Voir en français

An exabyte of disk storage at CERN

CERN disk storage capacity passes the threshold of one million terabytes of disk space

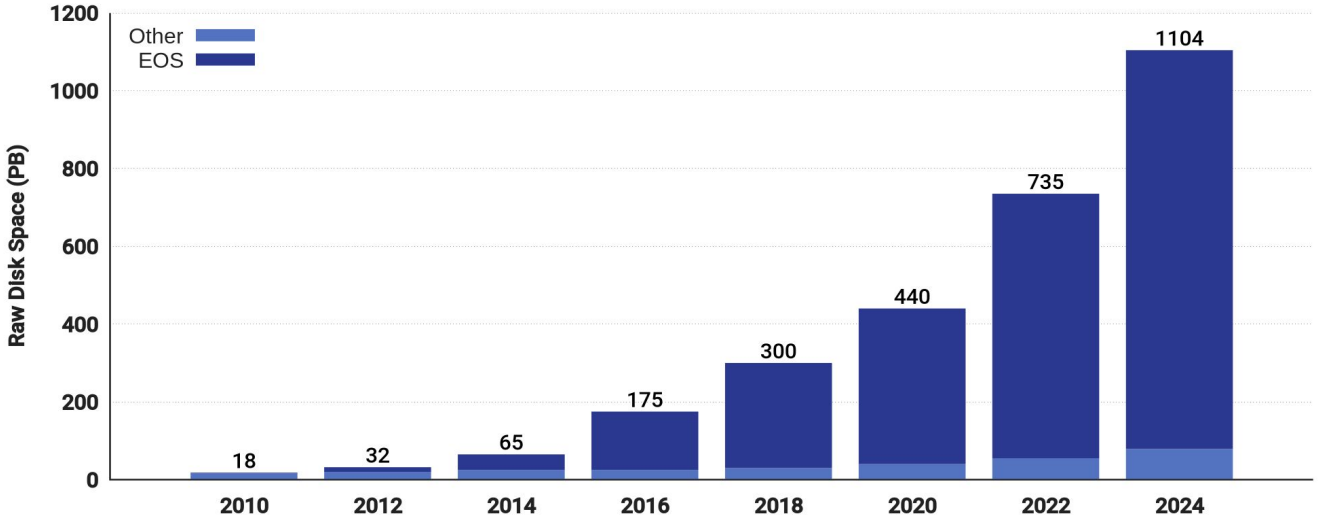
29 SEPTEMBER, 2023 | By Tim Smith

A fraction of the 111 000 devices that form CERN's data storage capacity. (Image: CERN)

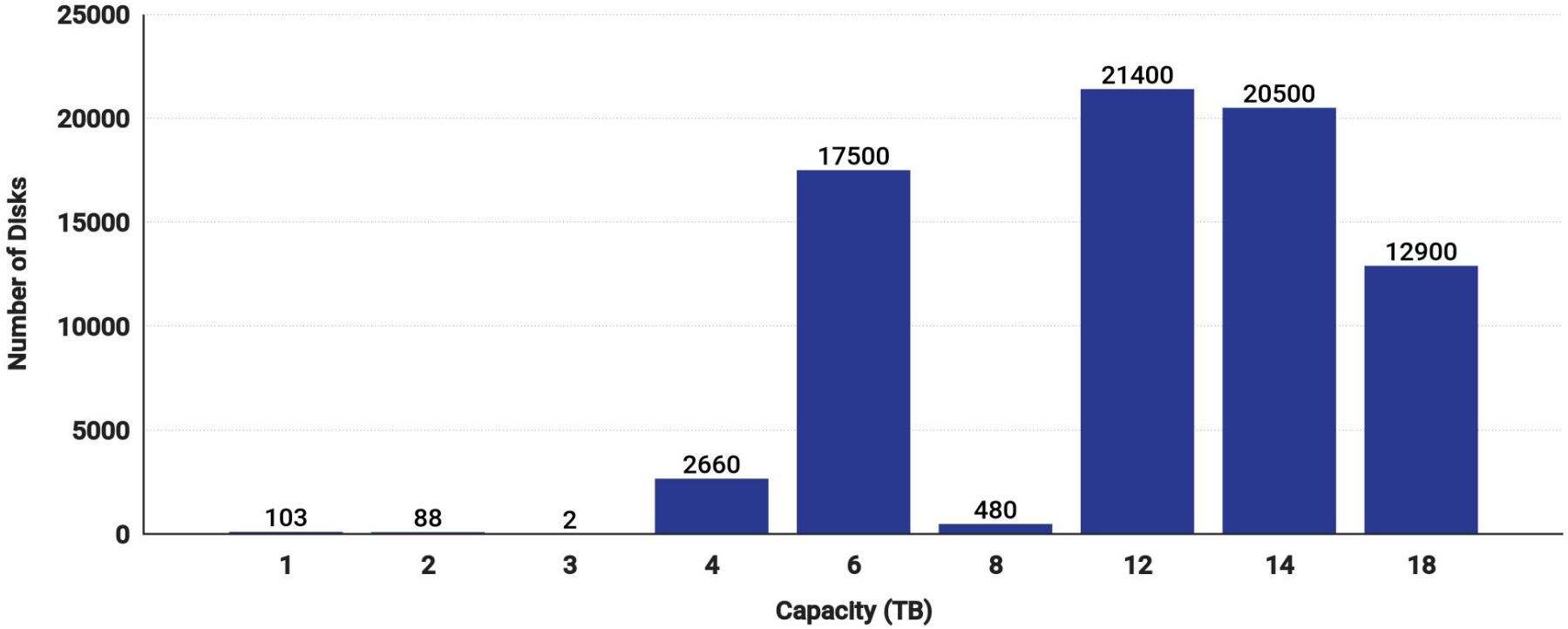
CERN's data store has now crossed the remarkable capacity threshold of one exabyte, meaning that CERN has one million terabytes of disk space ready for data!

CERN's data store not only serves LHC physics data, but also the whole spectrum of experiments and services needing online data management. This data capacity is provided using 111 000 devices, predominantly hard disks along with an increasing fraction of flash drives. Having such a large number of commodity devices means that component failures are common, so the store is built to be resilient, using different data replication methods. These disks, most of which are used to store physics data, are orchestrated by CERN's open-source software solution, EOS, which was created to meet the LHC's extreme computing requirements.

"We reached this new all-time record for CERN's storage infrastructure after capacity extensions for the upcoming LHC heavy-ion run," explains Andreas Peters, EOS project leader. "It is not just a celebration of data capacity, it is also a performance achievement, thanks to the reading rate of the combined data store crossing, for the first time, the one terabyte per second (1 TB/s) threshold."

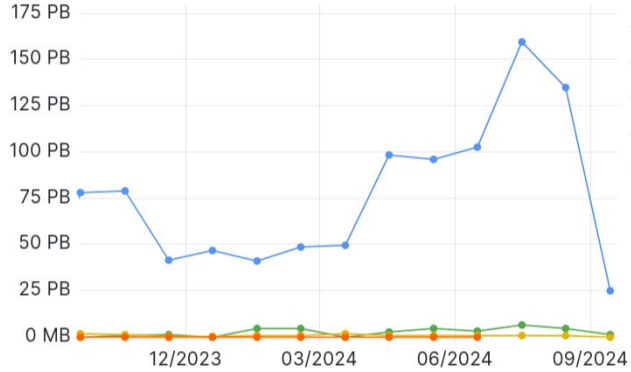


Mixture of Hardware Generations



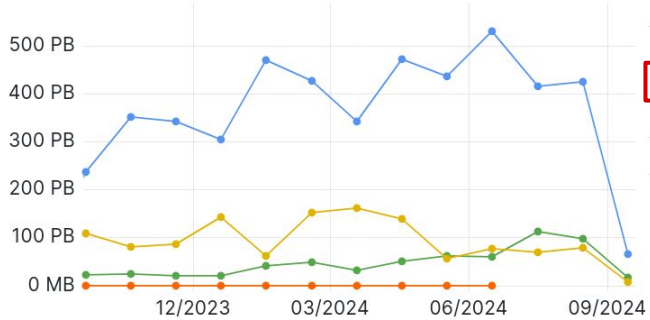
EOS Data Ingestion and Export Statistics (Past 12 months)

Ingestion per protocol

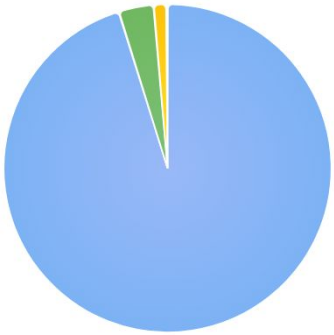


Name	Total
XROOTD	1.01 EB
HTTP	35.6 PB
FUSE	12.4 PB
GRIDFTP	350 TB

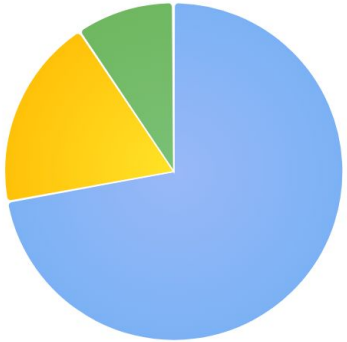
Export per protocol



Name	Total
XROOTD	4.86 EB
FUSE	1.24 EB
HTTP	624 PB
GRIDFTP	55.2 TB



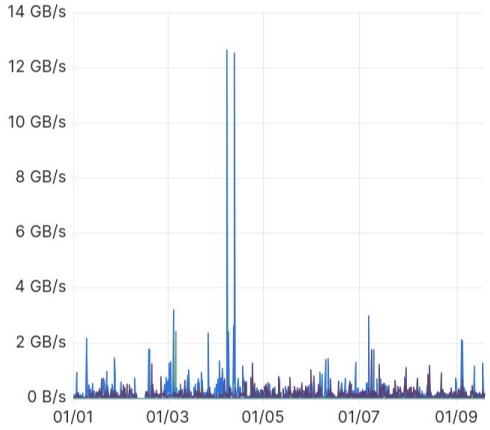
	Value	Percent
XROOTD	1.01 EB	95%
HTTP	35.6 PB	3%
FUSE	12.4 PB	1%
GRIDFTP	350 TB	0%



	Value	Percent
XROOTD	4.86 EB	72%
FUSE	1.24 EB	18%
HTTP	624 PB	9%
GRIDFTP	55.2 TB	0%

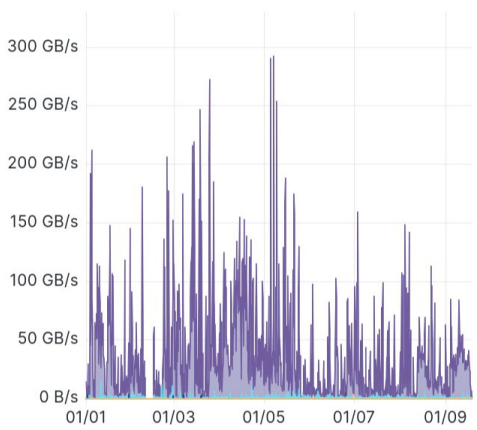
FUSE traffic vs Total Network Traffic in 2024

Fusex IN



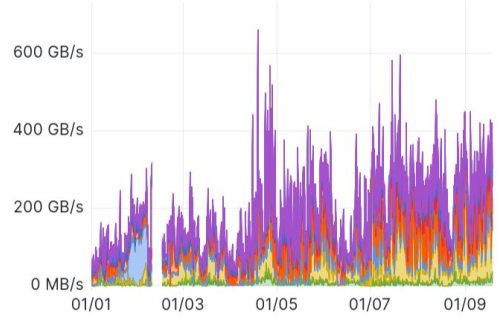
Name	Last *	Max
fuse::bi	4.87 MB/s	12.7 GB/s
fuse	2.61 kB/s	2.44 GB/s
fuse::lxplus	183 MB/s	1.77 GB/s
fuse::atljcluster	0 B/s	376 MB/s
fuse::fts3-streaming	0 B/s	367 MB/s
fuse::locmap	106 kB/s	282 MB/s
fuse::amssoc	137 B/s	151 MB/s
fuse::itgpt	0 B/s	149 MB/s
fuse::vocms	21.9 kB/s	130 MB/s
fuse::lbdt3	0 B/s	125 MB/s
fuse::voatlasmisc	0 B/s	74.3 MB/s
fuse::vocmsweb	0 B/s	64.4 MB/s
fuse::cmsp5	0 B/s	25.8 MB/s

Fusex OUT



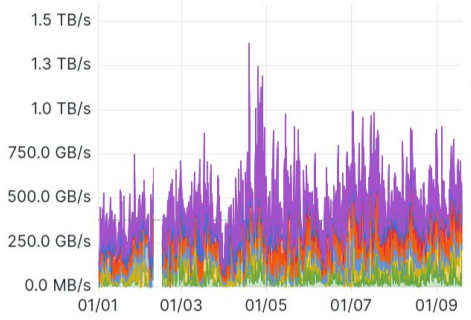
Name	Last	Max
fuse::bi	699 MB/s	292 GB/s
fuse::lxplus	1.18 GB/s	15.1 GB/s
fuse::lbdt3	0 B/s	5.34 GB/s
fuse	88.3 MB/s	5.05 GB/s
fuse::vocms	1.39 MB/s	3.54 GB/s
fuse::locmap	247 MB/s	3.01 GB/s
fuse::itgpt	0 B/s	461 MB/s
fuse::atljcluster	0 B/s	382 MB/s
fuse::samba	0 B/s	312 MB/s
fuse::na61	0 B/s	257 MB/s
fuse::swan	0 B/s	205 MB/s
fuse::openlab_systems	0 B/s	182 MB/s
fuse::amssoc	0 B/s	170 MB/s

Cluster Network Rates (in)



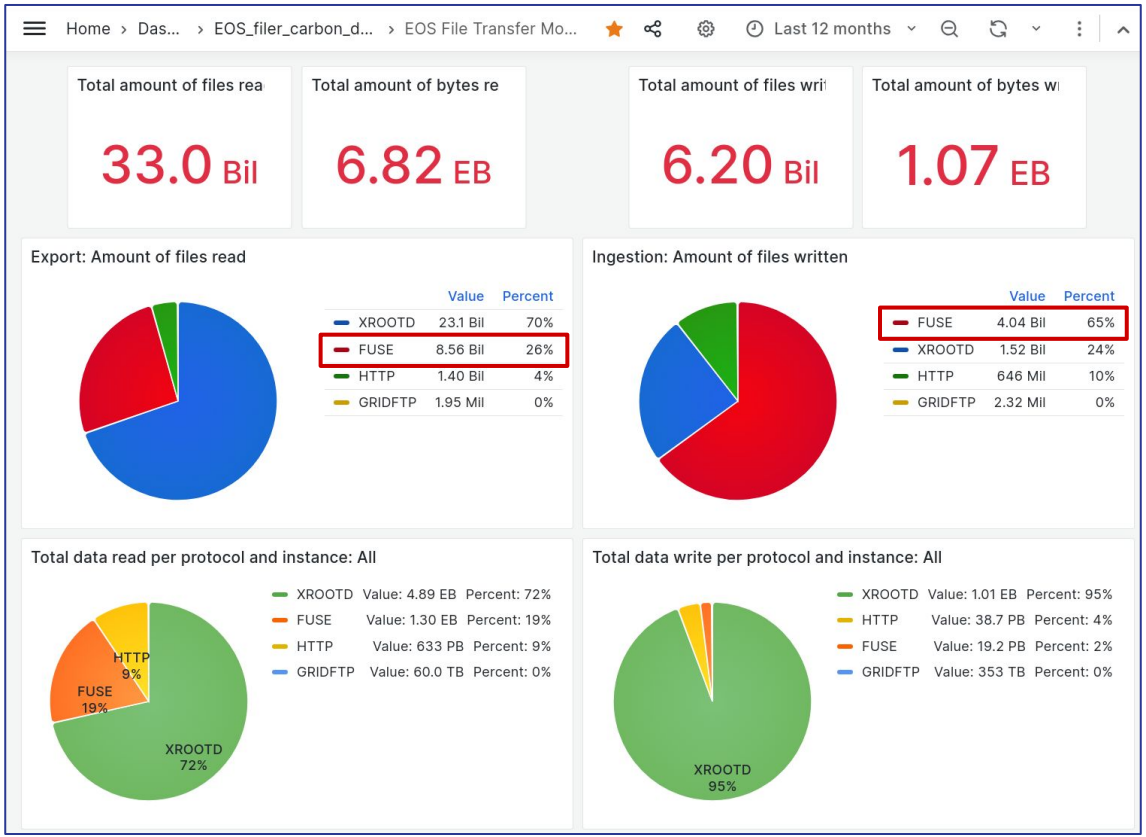
Name	Last *	Max	Mean
eosaliceo2	102 GB/s	485 GB/s	64 GB/s
eoscms	94 GB/s	223 GB/s	40 GB/s
eosatlas	42 GB/s	185 GB/s	34 GB/s
eosams02	10 GB/s	140 GB/s	13 GB/s
eosalice	39 GB/s	136 GB/s	9 GB/s
eoslhcb	40 GB/s	131 GB/s	14 GB/s
eospublic	6 GB/s	42 GB/s	11 GB/s

Cluster Network Rates (out)

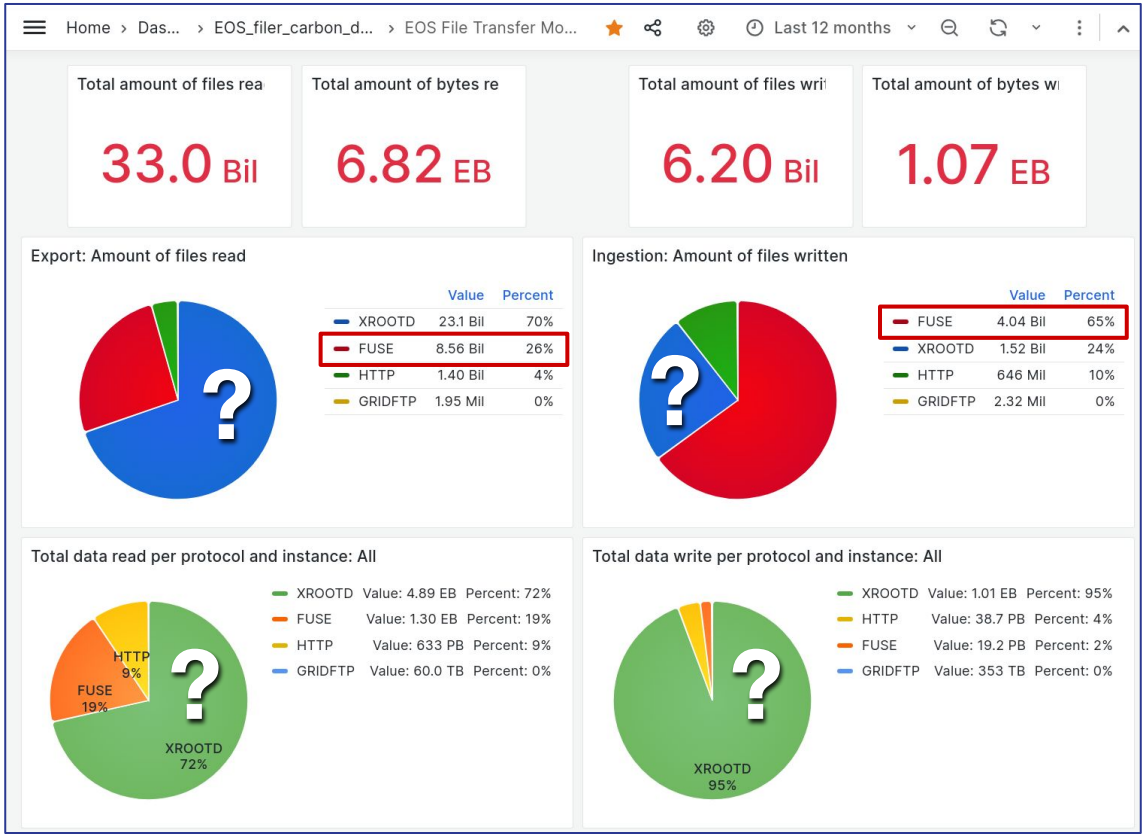


Name	Last *	Max	Mean
eosaliceo2	106.5 GB/s	856.1 GB/s	94.5 GB/s
eosatlas	37.4 GB/s	454.1 GB/s	61.2 GB/s
eosalice	175.6 GB/s	391.9 GB/s	75.3 GB/s
eoscms	149.7 GB/s	384.8 GB/s	93.1 GB/s
eospublic	34.6 GB/s	362.6 GB/s	48.9 GB/s
eoslhcb	79.6 GB/s	192.2 GB/s	26.9 GB/s
eosams02	12.9 GB/s	171.6 GB/s	44.3 GB/s

Data Ingestion and Export Statistics (Bytes vs Files)



Data Ingestion and Export Statistics (Bytes vs Files)



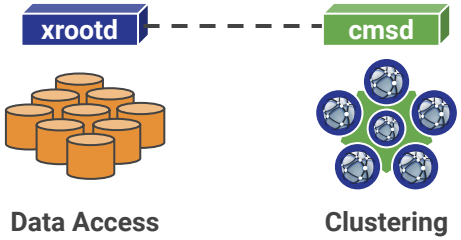


XRootD

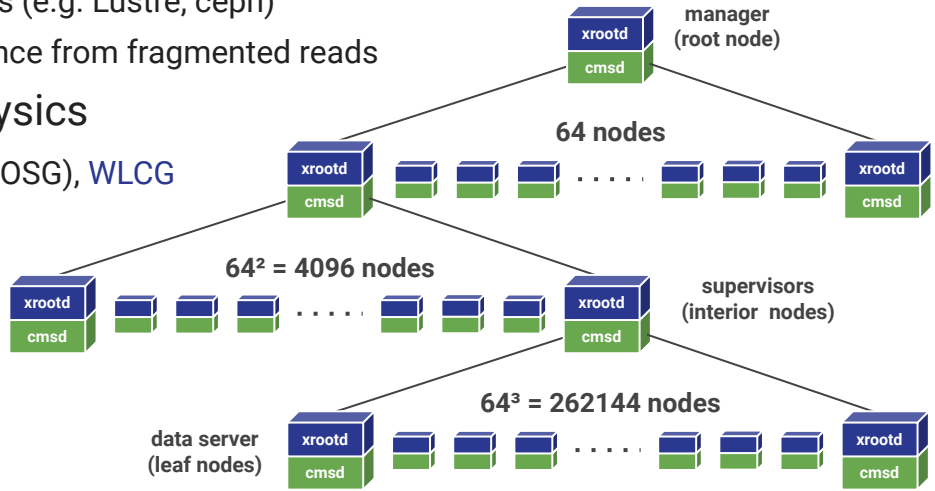
- XRootD is a system for scalable cluster data access
- Initially developed for BaBar experiment at SLAC (~2002)
  - The Next Generation ROOT File Server, A. B. Hanushevsky, A. Dorigo, F. Furano
- Written in C++, open source (LGPL + GPL)
- Available in EPEL and most Linux distributions
- You can think of XRootD as curl + nginx + varnish
  - Supports XRootD's own `root://` protocol as well as HTTP
  - `root://` is a stateful, POSIX-like protocol for remote data access
  - TLS (`roots://` and `https://`) support since XRootD 5.0
    - Supports TLS for control channel only, or control + data channel
- Can be configured also as proxy / caching server (XCache)
- Authentication via Kerberos, X509, shared secret, tokens
- Not a file system & not *just* for file systems

Requestid
kXR_auth
KXR_bind
kXR_chkpoint
kXR_chmod
kXR_close
KXR_dirlist
KXR_endsess
kXR_fattr
kXR_gpfile
kXR_locate
kXR_login
kXR_mkdir
kXR_mv
kXR_open
kXR_pgread
kXR_pgwrite
kXR_ping
kXR_prepare
kXR_protocol
kXR_query
kXR_read
kXR_readv
kXR_rm
kXR_rmdir
kXR_set
kXR_sigver
kXR_stat
kXR_stat
kXR_statx
kXR_sync
kXR_truncate
kXR_truncate
kXR_write
kXR_writev

XRootD



- XRootD clustering has many uses
  - Creating a uniform namespace, even though it is distributed
  - Load balancing and scaling in situations where all servers are similar
    - Proxy servers and caching servers (XCache)
    - Serving data from distributed filesystems (e.g. Lustre, ceph)
    - Ceph + XCache can avoid bad performance from fragmented reads
- Wide deployment across high energy physics
  - CMS AAA Data Federation, Open Science Grid (OSG), WLCG
- Highly adaptable plug-in architecture
  - If you can write a plug-in you can cluster it
  - Used by LSST (Vera C. Rubin Observatory)
    - LSST Qserv for clustered MySQL
    - https://inspirehep.net/literature/716175
- Extensive support for monitoring



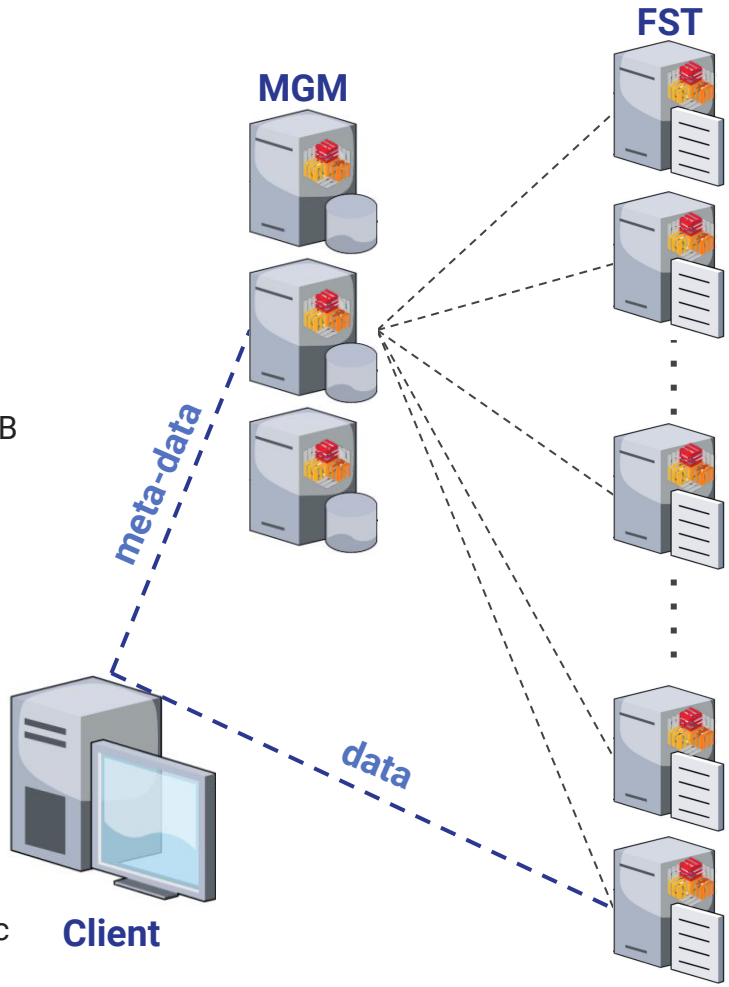


- Distributed filesystem built using the XRootD framework
  - XRootD provides the protocol, request redirections, third-party copy support
  - POSIX-like API, many authentication methods (Kerberos, X509, shared secrets, JWT tokens)
- Started in 2010 with the rationale to solve the physics analysis use-case for the LHC
- Extremely cost effective storage system (less than 1CHF / TB / month for EC 10+2 layout)
- Efficient sharing of resources with thousands of users across the grid
  - Quotas, ACLs for sharing, QoS features (data and metadata)
- Remotely accessible storage infrastructure
  - High energy physics distributed computing infrastructure with over 150 sites
  - FUSE mount provides convenient, familiar interface for users (batch, but also laptop)
- Storage system suitable for physics analysis use cases and data formats
  - ROOT columnar data format, heavy usage of vector reads for physics analysis
- Integration with OwnCloud web interface, sync & share, samba:



EOS Architecture

- Namespace server (MGM)
  - Scale up
  - Typically use 3 nodes (1 active, 2 standby)
  - Namespace persistency with key-value store (QuarkDB)
    - Developed at CERN, REDIS protocol, based on RocksDB
    - Separate process, usually runs along with the MGM
  - Namespace cache in RAM to reduce latency
- File storage servers (FST)
  - Scale out
  - Typically ~100–150 nodes
  - Based on cheap JBODs, RAIN layout
  - Multiple replica or erasure coding layouts supported
- Multiple instances with independent namespaces
  - /eos/{alice,atlas,cms,lhcb,public}, /eos/user, /eos/project, etc



EOS Features Needed by LHC Experiments

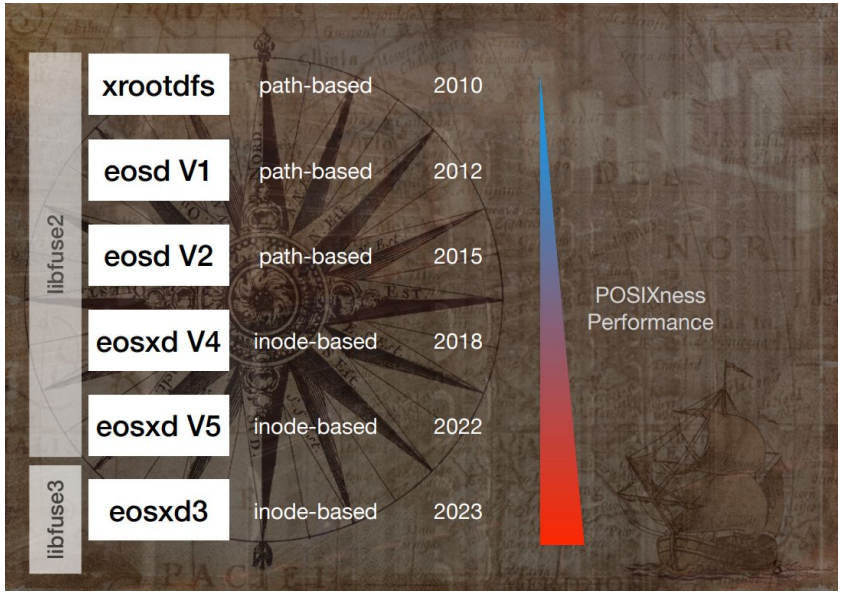
- Throttling
  - Limit rate of meta-data operations
  - Limit bandwidth per user, group, or application
  - Limit number of threads used by clients in the meta-data service
- Global limit on thread pool size and file descriptors
  - Run with at most N threads for all users, use at most N descriptors
- File placement policies
  - Layout (replica or erasure coding)
  - Physical hardware to use (HDD or SSD)
  - Geographic location (put EC pieces on separate nodes or groups)
- Conversion and Cleanup policies
  - Automatically convert large files from replica to EC layout or move to HDD
  - Periodically cleanup of files and/or empty directories from namespace



EOS and FUSE

- Origins with XRootD's xrootdfs
- Mostly libfuse2 client used in production
  - Roughly 30k active clients on average
- Libfuse3 used by SAMBA gateways only
- Issues in the past due to FUSE notification mechanisms
  - Invalidation, lookup, forget
  - Processes ending up in D state due to locking
- Uses XRootD protocol underneath
- Operations broadcast to clients
  - With a limit so as to not overload the server
- Local journal for writes
- First 256k of each file cached on open

History of EOS FUSE client



EOS and FUSE (2)

- Some problems we've encountered
  - Inode invalidation suppressed when write-back cache is enabled
    - Causes file sizes to become stale when another client changes a file (more information)
  - Problems seen on Alma Linux 9 due to mounting same filesystem more than once
- Areas of interest, suggestion for new features
  - Physics analysis makes heavy use of vector reads, FUSE breaks them up into multiple requests
    - Implemented hook in ROOT to bypass FUSE and read directly via XRootD when possible
  - Architecture in which meta-data caches could be shared between kernel and userspace would be desirable
- FUSE performance not a problem (bottlenecks are addressable on server side in EOS)
- Exploring ideas with FUSE passthrough (eoscfsd, add authentication to other backends)



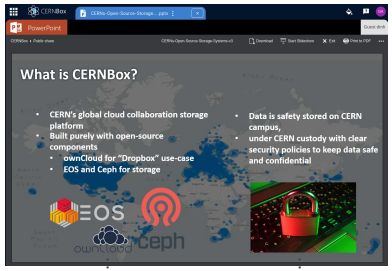


CernVM File system

- **C**ERN **V**irtual **M**achine **F**ile **S**ystem
- FUSE HTTP-based network read-only filesystem
- Originally developed to decouple experiment software from base OS
- Solution to distribute 1 TB per night of integration builds to the computing grid
- Origins as FUSE frontend to GrowFS using parrot system call interception
- Namespace stored in SQLite, content-addressed storage for data deduplication
- More recently being used to distribute “unpacked” container images
  - Use files directly from CVMFS, download and cache only what you need
- Software distribution on HPCs
  - Gentoo Prefix, ComputeCanada, EESSI



Storage Overview at CERN



CERNBox

Office 365

CodiMD

SWAN

ROOT Data Analysis Framework

XRootD

zenodo

CDS

CERN OpenData

RUCIO

FTS

indico

openstack

restic

CernVM File system

SAMBA

Sync & Share	Web Applications	Physics Analysis	FUSE

EOS

XRootD

AFS	Backup	Cloud

S3	RBD	Ceph FS

ceph

References and Further Information



Website

https://xroot.slac.stanford.edu

GitHub

https://github.com/xrootd/xrootd

XRootD/FTS Workshop (9–13 Sep 24)

https://indico.cern.ch/event/1386888



Website

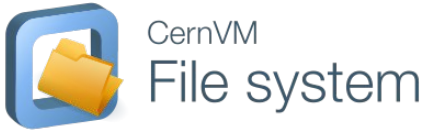
https://eos.web.cern.ch

GitHub

https://github.com/cern-eos/eos

EOS Workshop (14–15 Mar 24)

https://indico.cern.ch/event/1353101



Website

http://cernvm.cern.ch

GitHub

https://github.com/cvmfs/cvmfs

CVMFS Workshop (16 – 18 Sept 24)

https://indico.cern.ch/event/1347727


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YEARS / ANS CERN
Thank you!